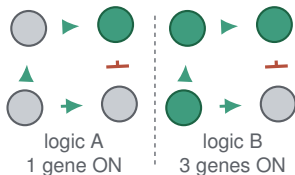


Logic over wiring: the model reads local Boolean logic, not just the wiring.

TRAIN DATA

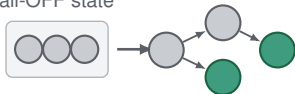
Synthetic Boolean nets ($n \leq 16$)

Same wiring, different logic
→ **different reachability**



Exact oracle

all-OFF state



→ per-node reachability labels

→ activation —| inhibition

MODEL

Sign-aware, size-agnostic GNN

positive
regulators



negative
regulators



+ channel

− channel

node update
(combine + and −)

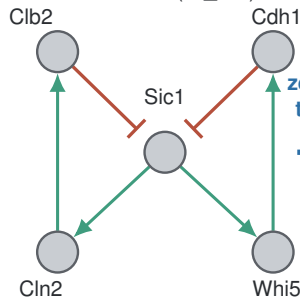
node features: logic + signs

trained on synthetic logic only

● reachable / ON

TRANSFER

Real GRNs ($n \leq 70$)

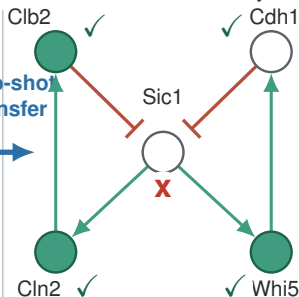


zero-shot
transfer

Exact check
symbolic BDD, validation only

OUTPUT

Predicted activatability



Coloured by prediction:

- activatable
- not activatable
- ✓ matches exact oracle
- ✗ differs from oracle

AUROC 0.91 [0.88–0.93]